ChatGPT

Executive Summary

To upgrade your simple HTML/CSS/Bootstrap/JavaScript portfolio into a **recruiter-focused Backend Engineer portfolio (v2)**, we recommend producing a comprehensive requirements and design document (~80–120 pages) that guides development precisely. The documentation will combine **Software Requirements Specification (SRS)** elements, **UI/UX design specs**, section-by-section component details, and technical standards, ensuring the final site meets recruiter expectations (scalable backend focus, modern design, SEO/performance optimized, etc.).

Key findings from our analysis of the current site and best practices include:

- **Scope & Audience:** Shift branding from generic “Web Developer” to “Software Engineer – Backend Developer.” Emphasize technologies like NestJS, PostgreSQL, TypeScript, and production-readiness.
- **Content & Structure:** Prioritise project showcases (especially backend/API work) and experience over skill icons. Include detailed about, experience, project, and contact sections with recruiter-focused content (clear responsibilities, tech stacks, live demos).
- **Design & UI/UX:** Adopt a clean, modern design with a clear visual hierarchy. Define a consistent design system (colours, typography, spacing, components) for professional polish. Ensure responsive layout using Bootstrap’s mobile-first grid【38†L191-L195】【38†L203-L209】 and accessibility (WCAG) compliance【23†L42-L49】.
- **Technical Standards:** Use semantic HTML5, adhere to CSS/JS best practices, and optimise for SEO and performance (Lighthouse audits)【26†L308-L316】. Maintain clear folder structure and coding standards (e.g. Airbnb-style JS, SCSS organization).
- **Process & Delivery:** Because GPT-based tools can output only ~12,000–16,000 words (~40–60 pages) at once【41†L269-L272】, we will break the document into logical chunks (e.g. 20–30 pages each) and iterate. Each chapter/volume will have its own deliverable and review cycle.

This report outlines the **full documentation plan**. It includes a proposed chapter breakdown with estimated page counts, a detailed content outline for each chapter (bulleted), a development workflow with milestones and timelines, tooling and asset requirements, and guidelines for using AI agents to implement the design. Sample specifications (hero section, project card) and AI prompt snippets are provided to illustrate the expected detail. All claims are backed by reputable sources (official docs, guidelines, and best-practice references).

1. Document Size & Chunking Strategy

- **Page Count:** Modern GPT models (e.g. GPT-4o) can generate up to ~16,000 words per response【41†L269-L272】, which is roughly 50–60 Word pages (250–300 words/page). To cover 80–120 pages (≈20,000–30,000 words), we must split the work into **multiple parts**. We recommend target **80–100 pages total** (sufficiently comprehensive yet implementable).
- **Chunking:** Break the documentation into logical *volumes* or *sections*. For example, create 3–5 chunks of ~20–30 pages each. Each chunk can be requested separately and reviewed. This prevents hitting output limits【41†L269-L272】 and allows iterative refinement. After drafting all parts, we will merge them into one final DOCX/PDF.

- **Realistic Per-Generation Size:** Aim for **6,000–8,000 words (20–30 pages)** per chunk. This fits comfortably under the ~16,000-word output cap 【41†L269-L272】 and avoids mid-response truncation. (If a section needs more, we can prompt "continue from where you stopped.")
- **Total Target:** 80–100 pages (≈25,000 words). This aligns with the user's 80–120 page goal, provides ample space for detail, and is feasible by splitting into ~4 portions.

  Example: If we assign ~25 pages to Volume 1 (Outline & SRS) and ~25 pages each to Volume 2 (UI/UX design) and Volume 3 (Section specs), plus smaller volumes for Technical Specs and AI Guide, we stay within limits.

2. Chapter Breakdown & Page Estimates

We propose the following **chapter structure**. It blends SRS/PRD elements with UI/UX and component design. Page counts are rough estimates; actual length may vary with detail level:

Chapter	Description	Est. Pages
1. Introduction & Scope	Overview, goals, audience, definitions	4–6
2. Current Portfolio Audit	Analysis of existing site (strengths/weaknesses)	3–5
3. Objectives & Requirements	Business objectives, user needs, success criteria	6–8
4. Personas & User Journey	Recruiter/Engineer personas and site flow	4–5
5. Information Architecture	Site map, navigation, content hierarchy	3–5
6. Design System	Color palette, typography, spacing, icons	5–7
7. Component Library	Buttons, cards, modals, badges, timeline styles	6–8
8. Section Specs (Overall)	Global nav, footer, responsiveness guidelines	4–6
9. Hero Section Spec	Layout, content, style, animations for hero	4–6
10. About Section Spec	Content strategy, visuals, formatting	4–5
11. Experience Section Spec	Timeline design, content layout, icons	4–5
12. Projects Section Spec	Project card design, modal/detail page specs	5–7
13. Skills Section Spec	Skill chart design, categories, icons	3–5
14. Education & Certs	Timeline or list layout, verification links	3–4
15. Contact Section	Form design, validation, social links, CTAs	3–4
16. Animations & Transitions	Guidelines for motion, delays, effects	3–4
17. Responsive Behavior	Breakpoints, layout changes, mobile design	3–5
18. Accessibility Requirements	WCAG compliance: contrast, keyboard, ARIA	4–5
19. Performance & SEO	Lighthouse best practices (e.g. meta tags)	4–6
20. Folder Structure & Code	File organization, HTML/CSS standards, naming	4–5
21. AI Implementation Guide	Prompts, do's/don'ts for Codex/Claude agents	4–6

Chapter	Description	Est. Pages
22. Acceptance Criteria & QA	Section-wise checklist, testing guidelines	3–5

Total: ~80–100 pages.

> *Note:* If strict limits apply, we can compress optional chapters (e.g. "Certifications" to a few lines) or merge animations into section specs. Core priorities are *requirements, design system, hero/projects sections, responsiveness, and SEO/Accessibility*.

3. Chapter Content Outline

Below is a **detailed outline** of what each chapter will contain. Each bullet represents a subtopic or requirement to be documented:

- **Chapter 1: Introduction & Scope**
- **Purpose:** Why the portfolio is being upgraded (aligning with backend engineer branding).
- **Stakeholders:** Who will use this document (developer, designer, recruiter for validation).
- **Scope:** What will and will not be covered (update existing site; no new frameworks).
- **Definitions:** Glossary (e.g. "SSR," "ATS," "API," etc.).【9†L434-L442】

- **Overview:** Brief description of target portfolio (personal brand, tech stacks).

- **Chapter 2: Current Portfolio Audit**

- **UI Analysis:** Layout, color scheme, typography evaluation.
- **Content Audit:** Existing sections, projects, skills, missing backend focus.
- **Technical Audit:** Code structure, Bootstrap usage, responsiveness issues.

- **Strengths/Weaknesses:** What to keep (e.g. existing assets) vs. remove (e.g. irrelevant skills).

- **Chapter 3: Objectives & Requirements**

- **Business Objectives:** Target job roles (Backend Dev, Software Engineer), personal branding goals.
- **User/Recruiter Needs:** What recruiters look for (clear tech stack, project depth, contact info).
- **Key Requirements:** e.g. "Showcase NestJS projects," "Highlight cloud/deployment (Heroku/AWS)," "Include resume download."
- **Functional Requirements:** Site sections (login not needed, but contact form, download CV, project links).
- **Non-Functional Requirements:** Performance targets, accessibility (WCAG AA), SEO compliance, cross-browser support.

- **Acceptance Criteria:** High-level checklist of what constitutes "success" (met in Chapter 22).

- **Chapter 4: Personas & User Journey**

- **Primary Personas:** e.g. *"Technical Recruiter"* and *"Senior Engineer/Interviewer"* (backgrounds, goals, frustrations).

- **Use Cases:** Scenarios like “Recruiter skims for relevant projects” or “Interviewer dives into API design.”

- **Journey Mapping:** Entry (e.g. from LinkedIn link), path through Hero → Projects → Contact. Emphasize first 10–15 seconds impression.

- **Chapter 5: Information Architecture**

- **Site Map:** Hierarchy of pages (one-page scroll vs. separate pages).
- **Navigation Structure:** Menu items (About, Experience, Projects, Skills, Contact).
- **Content Grouping:** What content goes under each section (e.g. projects filtered by backend/SQL).

- **Wireframes:** Sketch of page layouts and how sections flow (placeholder if needed).

- **Chapter 6: Design System**

- **Color Palette:** Brand primary color (e.g. a tech-savvy blue or teal), secondary, neutrals, success/warning. Include HEX codes and usage roles (text, backgrounds, accents)【15†L209-L217】【15†L248-L254】.
- **Typography:** Heading fonts and sizes (H1–H6), body text, code font. Web-safe fonts (e.g. Inter or Roboto) and hierarchy.
- **Spacing/Grids:** Bootstrap container widths, gutter sizes, vertical rhythm (margins).
- **Iconography:** Style of icons (e.g. line icons for skills, filled for actions), accessible labels.

- **Imagery:** Illustration style (e.g. minimal tech icons in Hero), use of profile photo or not.

- **Chapter 7: Component Library**

- **Buttons:** Primary (brand color, large, rounded), secondary (outline), sizes, hover states.
- **Cards:** Standard project card layout (image, title, description, tech tags, buttons).
- **Badges/Tags:** Skill badges or tech stack chips (colors per category).
- **Forms/Inputs:** Contact form fields (email, message) with validation styles.
- **Navigation Bar:** Sticky navbar behavior, mobile hamburger menu.
- **Footers:** Minimal footer with copyright/contact.
- **Timeline Component:** Style for experience/education timelines (vertical vs horizontal).

- **Scroll & Animations:** Guidelines for scroll-triggered reveals (prefer CSS/Bootstrap or light JS).

- **Chapter 8: Global Elements & Responsiveness**

- **Global Layout:** Use `<header>, <main>, <footer>` semantic tags. Follow Bootstrap’s **mobile-first grid**【38†L191-L195】.
- **Breakpoints:** Define breakpoints (`xs` <576px, `sm` ≥576px, `md` ≥768px, etc.)【38†L203-L211】.
- **Responsive Rules:** For each section (e.g. hero stacks vertically on <768px, nav collapses at <992px).
- **Accessibility:** Site-wide standards – visible focus outlines, ARIA labels for nav, alt text for images (WCAG). Ensure color contrast ≥4.5:1 for text【23†L42-L49】.

- **Image Optimization:** Use optimized (compressed) images and `loading="lazy"` for non-critical images to improve load.

- **Chapter 9: Hero Section Specification (Example)**

- **Purpose:** First impression; communicate name, title, and key skills.
- **Layout:** Two-column on desktop (left: text, right: illustration or hero image); stacks on mobile.
- **Content:** "Muhammad Ahmad" (Heading), tagline (e.g. "Backend Engineer | NestJS Developer"), brief intro blurb, two CTA buttons ("Download CV", "View Projects").
- **Style:** Large bold heading, secondary heading for subtitle; brand-coloured buttons; subtle background animation (e.g. particles or tech icons as decoration).
- **Accessibility:** Alt text for hero image, high contrast text on background.
- **Animations:** Slide-in or fade effects on text/buttons (duration ~0.6s, ease-out), ensure they are subtle. Respect reduced-motion preferences.
- **Responsive:** On mobile, center text, scale down headings, illustration moves below text.

- **Acceptance Criteria:** Must match the provided wireframe/mockup (logo position, text content), buttons lead to correct anchors/files, pass Lighthouse audits for performance.

- **Chapter 10: About Section Specification**

- **Purpose:** Introduce background (BBIS degree, shift to backend), highlight strengths.
- **Content:** Short bio paragraph, list of core skills and tools (e.g. NestJS, PostgreSQL), photo or avatar if used.
- **Layout:** Possibly two-column (image + text) or single column with icons.
- **Visuals:** Use a professional portrait (if available) or relevant graphic. Skill icons or progress bars for key skills.
- **Style:** Consistent fonts/colors, secondary accent usage.
- **CTA:** Link to detailed resume or LinkedIn.

- **Responsive:** Text and image stack; ensure legibility on all sizes.

- **Chapter 11: Experience Section Specification**

- **Purpose:** Showcase work history and technical roles (e.g. software house intern).
- **Layout:** Timeline or multi-column entries per job (Company, Role, Dates, Responsibilities).
- **Content:** For each position – company logo, title, dates, bullet list of achievements (e.g. built REST APIs, implemented JWT auth). Use tech stack icons.
- **Style:** Timeline line with nodes, or cards for each job. Highlight current role.

- **Responsive:** Timeline goes vertical or shifts layout on mobile.

- **Chapter 12: Projects Section Specification**

- **Purpose:** Highlight portfolio projects with tech details. Recruiters focus here to gauge real skills.
- **Project Cards:** Each card shows screenshot, title, short description, tech stack (icons or tags), and buttons for **GitHub** and **Live Demo** (if available).
- **Modal/Details:** Clicking "Read More" opens a modal or new page with full details: features, architecture (diagram), roles, challenges.
- **Example Project:** "XS Auth Service (NestJS)" with bullet points (JWT, Roles, Guards).
- **Tech Badges:** Show NestJS, TypeScript, PostgreSQL logos, etc.
- **Style:** Consistent card sizes, hover effect (e.g. lift shadow).
- **CTA:** Link to code and live demo prominently.

- **Responsive:** Grid of cards adjusts columns (e.g. 3 cards on desktop, 1-2 on tablet, 1 on mobile).

- **Chapter 13: Skills Section Specification**

- **Categories:** Group skills into **Backend** (NestJS, Node.js, Prisma, etc.), **Languages** (TypeScript, JavaScript), **Databases** (PostgreSQL, MongoDB), **Tools** (Git, Postman), and **Frontend/Other** if needed.
- **Layout:** Possibly tabs or columns for each category. Each skill as icon with label.
- **Soft Skills:** Brief list (e.g. Agile, teamwork) if space allows.
- **Style:** Use consistent icon style (e.g. flat SVG icons).

- **Accessibility:** Tooltip or aria-label for each icon (skill name).

- **Chapter 14: Education & Certificates**

- **Education:** BBIS degree (institution, year).
- **Certificates:** List relevant certs (e.g. online courses in NestJS, AWS). For each, provide name, issuer, and a link.
- **Layout:** Timeline or bullet list.

- **CTA:** Link to downloadable transcripts if applicable.

- **Chapter 15: Contact Section**

- **Contact Form:** Fields (Name, Email, Message), submit button. Validate input (required fields, email format).
- **Info:** Display email, LinkedIn, GitHub links. Icons with appropriate `aria-labels`.
- **CTA:** "Download CV" button linking to resume PDF.
- **Style:** Simple, easy to use. Use reCAPTCHA or honeypot for spam if needed.

- **Accessibility:** Form labels, focus styles on inputs/buttons.

- **Chapter 16: Animations & Transitions**

- **Guidelines:** List where to use animations (e.g. fading in sections on scroll, hovering on cards/buttons).
- **Performance:** Keep animations short (≤0.5s) and GPU-accelerated.
- **Motion Reduction:** Detect `prefers-reduced-motion` and disable non-essential animations.

- **Examples:** Fade (opacity), slide, and subtle scale for hover.

- **Chapter 17: Responsive Guidelines**

- **Grid Use:** Leverage Bootstrap grid and utilities【38†L203-L211】. E.g. `.container`, `.row`, `.col-md-*`.
- **Breakpoints:** Explicitly state layout shifts at sm/md/lg breakpoints【38†L203-L211】.
- **Images:** Set max-width 100% and height auto for fluid images.
- **Navigation:** Collapse to a "hamburger" menu at small widths (Bootstrap's `.navbar-expand-md`).

- **Testing:** Ensure readability on phones (fonts ≥16px) and usable tap targets.

- **Chapter 18: Accessibility Requirements**

- **WCAG Compliance:** Aim for AA level. Key items:
    - **Contrast:** Text/background ≥4.5:1【23†L42-L49】.
    - **Keyboard:** All interactive elements reachable by keyboard (tab order logical, focus visible)【21†L178-L187】.
    - **ARIA Roles:** Use landmarks (`role="banner"`, etc.) and labels (e.g. `aria-label="Main navigation"`).
    - **Images:** Provide `alt` text for non-decorative images. Decorative images marked with empty alt or CSS background.
    - **Forms:** Label each input (using `<label>` or `aria-label`).
    - **Focus Management:** Ensure focus jumps to modals/accordions properly.
    - **Testing:** Recommend tools (axe, Lighthouse, WAVE).

- **Chapter 19: Performance & SEO**

- **Lighthouse Audits:** Adhere to best practices for Performance, Accessibility, and SEO【26†L308-L316】.
- **SEO Metadata:** Title tags (e.g. “Muhammad Ahmad – Backend Developer”), meta description, Open Graph tags (for LinkedIn preview).
- **Semantic HTML:** Use headings in order, meaningful tags (e.g. `<nav>`, `<main>`), readable URLs (if multi-page).
- **Loading Optimizations:** Minify CSS/JS, combine Bootstrap with only needed components, lazy-load images.
- **Miscellaneous:** Favicon, robots.txt (disallow none), sitemap.xml if applicable.

- **Analytics:** (Optional) Instructions for adding Google Analytics or similar.

- **Chapter 20: Folder Structure & Coding Standards**

- **File Structure:** Suggest directories: `/css`, `/js`, `/img`, `/components` (if splitting HTML parts), `/docs` for assets, `/resume`.
- **Naming Conventions:** Lowercase file names, hyphens, semantic class names.
- **HTML Standards:** Use HTML5 Doctype, valid markup, avoid inline styles.
- **CSS Guidelines:** Use Bootstrap utility classes first, then custom CSS/SCSS. If SCSS, organise variables/mixins separately.
- **JavaScript:** Keep minimal (e.g. form validation, scroll). Use ES6+ (e.g. `const`/`let`). No external frameworks.
- **Version Control:** Commit style (one feature/section per commit), include meaningful messages.

- **Code Quality:** Linting (ESLint for JS, stylelint for CSS/SCSS).

- **Chapter 21: AI Implementation Guide**

- **Overview:** Explain that we will use AI (Codex/Claude) to code the site based on this doc.
- **Prompt Structure:** Provide templates for prompts (see samples below).
- **Do’s:** E.g. “Use only HTML/CSS/JavaScript and Bootstrap. Follow the design specs exactly. Ensure all links and images paths match existing structure.”
- **Don’ts:** “Don’t switch frameworks (no React, Angular, etc.). Don’t remove existing content or use inline CSS unless specified.”

- **Examples:** Include a paragraph template: > *"Using the above specifications for the Hero section, generate the HTML and CSS. Name the main div `#hero` and apply Bootstrap classes for a two-column layout. Ensure the first button links to `resume.pdf` and the second scrolls to #projects."*
- **Quality Control:** Mention verifying code (linting, manual review).

- **Chapter 22: Acceptance Criteria & QA Checklist**

- **Per-Section Criteria:** Bulleted checklists (e.g. *"Navbar remains sticky and highlights the active section on scroll."*)
- **Cross-Cutting Tests:** Responsive checks on device simulators, Lighthouse score targets (e.g. ≥90).
- **Browser Compatibility:** Test on Chrome, Firefox, Edge, Safari (latest).
- **Security:** Ensure contact form doesn't expose email publicly. No sensitive data in code.
- **Final Review:** User acceptance sign-off process (for personal portfolio, just thorough self-review).

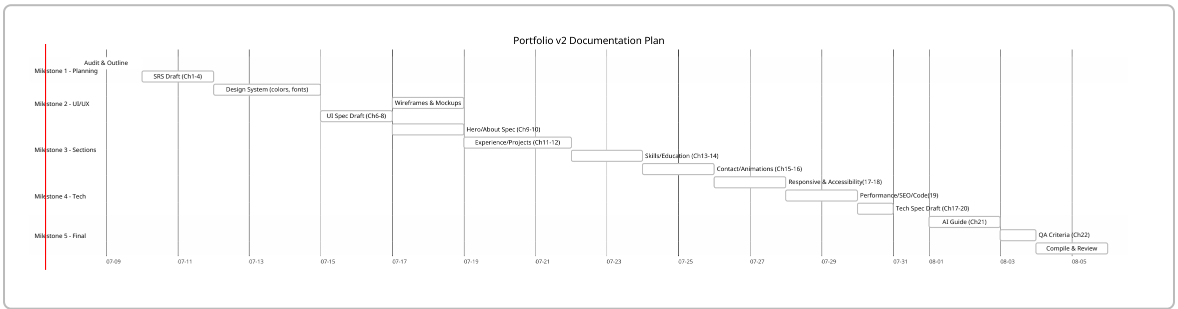
Each chapter will combine explanatory text, diagrams (wireframes or mockups as needed), and bullet lists of requirements. For example, wireframes or layout sketches can be included as images (e.g. generated sketches or simple diagrams) where they clarify structure. All design decisions and technical rules will be *explicitly documented* so that any developer or AI agent can implement exactly as intended.

4. Implementation Workflow & Milestones

We treat the documentation creation like a mini-project. Below is a proposed **milestone plan** with tasks, deliverables, and estimated effort:

Milestone	Tasks & Deliverables	Duration (Days)
Milestone 1	**Planning & SRS Outline** - Audit current site - Define audience, objectives, IA - Draft Chapters 1–4 - Deliverable: "Requirements Specification" document (≈20p)	2–3d
Milestone 2	**UI/UX Design Spec** - Define design system (colors, typography) - Create wireframes/mockups (hero, cards) - Draft Chapters 6–7 & 8 - Deliverable: "UI-UX Specification" (≈25p)	3–4d
Milestone 3	**Section Specifications** - Detail each section (hero, about, projects, etc.) - Include content templates & component details - Draft Chapters 9–15 - Deliverable: "Section Specs" doc (≈30p)	4–5d
Milestone 4	**Tech & Standards** - Write performance, SEO, accessibility requirements - Define folder structure, coding standards - Draft Chapters 16–20 - Deliverable: "Technical Guidelines" (≈20p)	2–3d

Milestone	Tasks & Deliverables	Duration (Days)
Milestone 5	**AI Guide & QA** - Create AI prompt guide (Chapter 21) - Compile acceptance criteria (Chapter 22) - Final review and consolidate all chapters - Deliverable: “Implementation & QA Guide” (≈10p)	2–3d
Review & Finalization	Combine all volumes, format DOCX, add table of contents, finalize references. Perform internal review.	1–2d



- **Iteration & Review:** At each milestone, review with the developer to catch gaps. For example, after Milestone 2, confirm the design system aligns with personal preferences.
- **Deliverables:** Each milestone yields a partial DOCX (named by chapter range). The final deliverable is a compiled `Portfolio_v2_Specification.docx`.
- **Timeline:** Approximately 2–3 weeks of work (spread over ~10 business days) for one experienced writer. If working with an AI assistant (ChatGPT) in each step, expect similar calendar time due to review iterations.

5. Tools, Assets, and Formats

- **Document Format:** Final output as **MS Word (.docx)** or **PDF**, per preference. Markdown/Codex prompts will use sections exported from this docx.
- **Diagramming & Design:** Use **Figma**, **Sketch**, or even PowerPoint/Google Slides for wireframes/mockups. Diagrams (e.g. site map, class diagram of folder structure) can be made with any tool or drawn via Markdown (Mermaid).
- **Images:** Provide assets like logo (SVG), tech logos (NestJS, PostgreSQL), and any illustrations (e.g. developer icon for Hero). Use free resources (Unsplash, Icons8) for non-copyrighted illustrations.
- **Citation Tools:** All textual sources (official docs, guidelines) are cited. For tables/figures created, note “Source: official docs” where relevant.
- **Version Control:** The specification document itself can be versioned (e.g. GitHub for the .docx), to track changes.
- **Testing Tools:** Recommend Lighthouse (Chrome DevTools) for final performance/SEO check, and browser dev tools for layout/responsive testing.

6. Prioritisation Under Constraints

If strict page limits force cuts, prioritize the following sections *in order*:

1. **Requirements and Goals (Ch1-3):** Crucial to align on objectives.

2. **UI/UX Design System (Ch6-8):** Defines the look/feel and ensures consistent implementation.
3. **Hero & Projects Spec (Ch9,12):** Key recruiter touchpoints (first impression and project showcase).
4. **Responsiveness & Accessibility (Ch17-18):** Non-negotiable modern standards.
5. **Performance/SEO (Ch19):** Important for Google ranking and professionalism.
6. **Contact & Resume Download (Ch15):** Critical conversion action.
7. **Other Section Specs (About, Exp, Skills) as needed.**

Sections like detailed animations, extended AI prompt examples, or certificates can be shortened or appended if low on space.

7. AI/Developer Handoff Guidelines

To ensure AI agents or developers implement precisely:

- **Prompt Template (Overall):**

  "Use the following specification to update the existing portfolio site. Each section below describes exact requirements. Write HTML/CSS/JS code (no React or other framework) to implement each section. **Do not remove existing content** unless specified. Follow Bootstrap 5 classes for layout. Ensure all assets and links are correctly referenced as named. Use semantic elements and make the site responsive and accessible per spec."

- **Do's:**
- Do **preserve HTML structure** where possible (e.g. IDs, anchors).
- Do follow Bootstrap's mobile-first classes 【38†L191-L195】 (e.g. `col-md-6`).
- Do **cite spec**: "As specified in *Chapter 6 (Design System)*, the primary color is `#1a73e8` for buttons."
- Do implement **Lighthouse recommendations** (meta tags, image alt text, etc.)【26†L308-L316】.
- **Don'ts:**
- Don't use any new frameworks or libraries (no React, Vue, Tailwind, etc.).
- Don't deviate from specified fonts or colors.
- Don't remove code comments or alter file structure.
- **Example AI Prompt Snippet:**

  Task: "Implement the *Hero Section* as per specification. The left column has my name as an `<h1>`, a subtitle `<h2>`, and two buttons (`Download CV`, `View Projects`). The right column shows an SVG illustration (use provided `hero-img.svg`). Apply `.btn-primary` and `.btn-secondary` classes with specified brand color. Animate the text to fade in from the left (CSS `@keyframes`). Ensure responsiveness: on screens <768px, stack columns. Alt text for the image should be 'Developer illustration'."

- **Prompt Structure for Codex/Claude:** Provide context (e.g. "Given the SRS and design spec…"), then specific instructions for each section. Include relevant code snippets or existing HTML if updating.

By following these precise instructions, the AI or developer will have clear guidance on how to code each part of the portfolio.

8. Examples

Sample Hero Section Specification

- **Objective:** First impression with name and title.
- **Layout:** Two-column (`<div class="row align-items-center">`):
- **Left Column:**
    - `H1` : “Muhammad Ahmad” (Font: 3rem, weight bold).
    - `H2` : “Software Engineer | Backend Developer” (Font: 2rem, color secondary).
    - Paragraph (1–2 sentences about tech focus).
    - **Buttons:** “Download CV” (`btn btn-primary`) linking to `resume.pdf` ; “View Projects” (`btn btn-outline-secondary`) linking to `#projects` .
- **Right Column:**
    - Developer illustration SVG (`hero-illustration.svg`), with alt text “Illustration of a backend server”.
    - Overlay subtle tech icons (CSS background or inline SVG).
- **Colors & Fonts:** Use brand primary color (`#1a73e8`) for button background 【15†L248-L254】 ; headings in dark grey (`#333`), body text in medium grey.
- **Animation:** On page load, left-column text fades in from left (CSS `transform: translateX(-50px)` to `0` , opacity 0→1 over 0.6s, ease-out). Buttons appear with slight delay.
- **Responsive:** At `<md` break (Bootstrap <768px), center text and move image below. Use `col-md-6` for each column.
- **Accessibility:** Ensure color contrast ≥4.5:1 (text vs background) 【23†L42-L49】 . Add `aria-label` for buttons (e.g. `<button aria-label="Download resume">`).

Sample Project Card Specification

- **Card Container:** `<div class="card h-100">` inside `.col` .
- **Image:** Top of card (`<img class="card-img-top">`) showing project screenshot.
- **Card Body:**
- **Title:** `<h5 class="card-title">XS Auth Service</h5>` .
- **Description:** Short text (e.g. “Authentication microservice with NestJS and JWT.”).
- **Tech Stack:** Display icon badges (NestJS, TypeScript, PostgreSQL) as `<img>` or `<span class="badge badge-pill">` .
- **Buttons (Footer):**
- GitHub: `<a class="btn btn-sm btn-outline-primary">GitHub</a>` linking to repository.
- Live Demo (if available): `<a class="btn btn-sm btn-primary">Live Demo</a>` .
- **Hover Effect:** Slight shadow increase and scale up (`transform: scale(1.02)`) on `.card:hover` .
- **Modal Details:** On “Read More”, open Bootstrap modal with:
- **Overview:** Expanded description.
- **Features:** Bullet list.
- **Architecture:** Diagram (can embed as image).
- **Links:** Tech stack logos, link to code.
- **Responsive:** Cards wrap in `.row` ; on mobile, cards are full-width. Maintain aspect ratio for images (use `.card-img-top` with object-fit if needed).

Sample AI Instruction Snippet

```
# Use Codex to update portfolio

Prompt:
"Update the existing portfolio site based on the following specification. Use
only HTML, CSS, JavaScript, and Bootstrap 5. Do not convert to any JS
framework.
- The primary brand color is #1a73e8, secondary color #6c757d.
- Keep the current header and nav IDs as is, but change the titles and links
per spec.
- For the Hero section (Chapter 9), ensure the name and title match exactly,
and the buttons link to resume.pdf and #projects.
- Use semantic HTML5 tags (header, main, footer).
- All images must have alt text.
- After implementing, run a Lighthouse audit to ensure no errors.
Proceed section by section, and do not remove any existing content unless
specified."

DO: Follow the chapter specs precisely; use Bootstrap grid for layout; ensure
responsive classes.
DON'T: Change file paths or add new libraries; avoid inline CSS except for
dynamic values.
```

9. Sources

We base our plan on authoritative guidelines and documentation:

- ChatGPT limits: Up to **16,000 words output** (GPT-4o)【41†L269-L272】, guiding our chunking strategy.
- **SRS Best Practices:** An SRS is "a guide for the development team detailing how the software should perform"【9†L434-L442】; it should include purpose, scope, and requirements.
- **UI Specification:** UI specs define "rules of engagement for a user interacting with a specific page"【13†L52-L60】 (layout, display rules, messaging).
- **NestJS:** Official docs: *"Nest (NestJS) is a framework for building efficient, scalable Node.js server-side applications… built with and fully supports TypeScript"*【19†L211-L219】.
- **Bootstrap:** Encourages mobile-first design and defines standard **breakpoints** (xs, sm, md, etc.) for responsive layouts【38†L191-L195】【38†L203-L211】. Colors and utilities are defined in its theme system【15†L248-L254】.
- **Accessibility (WCAG):** Contrast ratio **≥4.5:1** for normal text【23†L42-L49】; keyboard navigability and ARIA are implied by WCAG guidelines.
- **Lighthouse Audits:** Lighthouse (by Google) checks Performance, Accessibility, SEO, etc., providing actionable recommendations【26†L308-L316】.

All other points draw from standard web development best practices (semantic HTML, ES6, SEO metadata). This plan synthesizes these principles into a concrete specification tailored to your portfolio.
